The existence of quantised charge

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Consider the complex-valued massive scalar field Lagrangian:

$$\mathcal{L} = \partial_\mu \psi^* \partial^\mu \psi - M^2 \psi^* \psi \tag{1}$$

We seek to find a general form for ψ. Apply the Euler-Lagrange equation to obtain and equation of motion:

$$\frac{\partial \mathcal{L}}{\partial \psi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} \right) = 0 \tag{2}$$

Note that we can vary with respect to either ψ or ψ^*. Varying with respect to one will produce an equation of motion in the other. In general, that form is:

$$(\Box + m^2)\psi = 0 \tag{3}$$

A general plane wave of the following form solves this equation, called the Klein-Gordon equation:

$$\psi(x) = e^{ik\cdot x} \tag{4}$$

With the restriction:

$$k_\mu k^\mu = m^2 \tag{5}$$

And with:

$$k_\mu k^\mu = \omega^2 - \mathbf{p}^2 \tag{6}$$

Of course, expression (4) is a very specific solution, and as is usual with solutions to differential equations, a spectrum is required. Therefore, we can write:

$$\psi(x) = \int \frac{d^3x}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{p}}}} (a_{\mathbf{p}} e^{-ip\cdot x} + b^*_{\mathbf{p}} e^{ip\cdot x}) \tag{7}$$

For this to become a quantum field theory, we promote $a_{\mathbf{p}}$ and $b_{\mathbf{p}}$ to operators satisfying the following commutation relation (valid for both a and b)

$$[\hat{a}_{\mathbf{p}}, \hat{a}^\dagger_{\mathbf{q}}] = (2\pi)^3 \delta^3(\mathbf{p} - \mathbf{q}) \tag{8}$$

(Where conjugation a^* becomes Hermitian conjugation $\hat{a}^\dagger$) The canonical momentum is defined as:

$$\pi(x) = \frac{\partial \mathcal{L}}{\partial \dot{\psi}} \tag{9}$$

The Hamiltonian is the usual Legendre transform of the Lagrangian, given in this case as:

$$H = \int d^3x \, (\dot{\psi}\pi + \dot{\psi}^* \pi^* - \mathcal{L}) \tag{10}$$

After some rather hefty algebra (not too conceptually challenging, just messy) (also ignoring any delta functions evaluated as 0, since these represent zero point energy of the field) we are left with the nice expression:

$$\hat{H} = \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} (\hat{a}^\dagger_{\mathbf{p}} \hat{a}_{\mathbf{p}} + \hat{b}^\dagger_{\mathbf{p}} \hat{b}_{\mathbf{p}}) \tag{11}$$

Which follows somewhat intuitively due to the pairing of the raising and lowering operators being the number operator; giving you the total number of particles in that state. Therefore, this Hamiltonian tells you:

$$\int (\text{energy of state}) \times (\text{number in state})\, d\,(\text{all states}) \tag{12}$$

Defining the vacuum state as $\hat{a}_{\mathbf{p}} |0\rangle = 0$ and the momentum state as $|\mathbf{p}\rangle_a = \hat{a}^\dagger_{\mathbf{p}} |0\rangle$ we can perform a sanity check by finding the eigenvalues of $\hat{H}$ on a particle state $|\mathbf{p}\rangle_a$. Let us first find the commutator $[\hat{H}, \hat{a}^\dagger_{\mathbf{p}}]$:

$$[\hat{H}, \hat{a}^\dagger_{\mathbf{q}}] = \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} ([\hat{a}^\dagger_{\mathbf{p}} \hat{a}_{\mathbf{p}}, \hat{a}^\dagger_{\mathbf{q}}] + [\hat{b}^\dagger_{\mathbf{p}} \hat{b}_{\mathbf{p}}, \hat{a}^\dagger_{\mathbf{q}}]) \tag{13}$$

Which simplifies down to:

$$[\hat{H}, \hat{a}^\dagger_{\mathbf{q}}] = \int \frac{d^3p}{(2\pi)^3} \omega_{\mathbf{p}} (2\pi)^3 \delta(\mathbf{p} - \mathbf{q}) \hat{a}^\dagger_{\mathbf{p}} = \omega_{\mathbf{p}} \hat{a}^\dagger_{\mathbf{p}} \tag{14}$$

Therefore, to find $\hat{H} |\mathbf{p}\rangle_a = \hat{H} \hat{a}^\dagger_{\mathbf{p}} |0\rangle$, we utilise the above commutator: (and $[AB] = AB - BA$, of course)

$$\hat{H} \hat{a}^\dagger_{\mathbf{p}} |0\rangle = \left([\hat{H}, \hat{a}^\dagger_{\mathbf{p}}] + \hat{a}^\dagger_{\mathbf{p}} \hat{H} \right) |0\rangle \tag{15}$$

Since $\hat{H} |0\rangle$ is naturally 0, we are left with:

$$\hat{H} \hat{a}^\dagger_{\mathbf{p}} |0\rangle = \omega_{\mathbf{p}} \hat{a}^\dagger_{\mathbf{p}} |0\rangle = \omega_{\mathbf{p}} |\mathbf{p}\rangle \tag{16}$$

Which naturally follows. Now that we have a solid particle framework, we can apply some ideas from some of my previous writings. The Lagrangian we are working with is invariant under $U(1)$ transformations. We already know this implies the existence of the electromagnetic field, but now we will see what this conserved charge actually is. Let us consider the Noether current:

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} \delta\psi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi^*)} \delta\psi^* \tag{17}$$

We can logically reason that, for small α, $e^{i\alpha} \approx i\alpha$, so:

$$\delta x = i\alpha\psi \qquad \delta x^* = -i\alpha\psi^* \tag{18}$$

So the Noether current becomes (factoring out the constant α:

$$j^\mu = i(\psi^* \partial^\mu \psi - \psi \partial^\mu \psi^*) \tag{19}$$

This yields the conserved charge:

$$Q = \int d^3x j^0 = \int d^3x (\psi^* \dot{\psi} - \psi \dot{\psi}^*) \tag{20}$$

Which when quantised, yields:

$$\hat{Q} = \int \frac{d^3p}{(2\pi)^3} (\hat{a}^\dagger_{\mathbf{p}} \hat{a}_{\mathbf{p}} - \hat{b}^\dagger_{\mathbf{p}} \hat{b}_{\mathbf{p}}) \tag{21}$$

This implies that the difference between the number of a particles and the number of b particles must stay the same. The net charge must stay the same. This becomes clearer if we enact $\hat{Q}$ on the a and b states. First, determine $[\hat{Q}, \hat{a}^\dagger_{\mathbf{a}}]$:

$$[\hat{Q}, \hat{a}^\dagger_{\mathbf{q}}] = \int \frac{d^3p}{(2\pi)^3} [\hat{a}^\dagger_{\mathbf{p}} \hat{a}_{\mathbf{p}}, \hat{a}^\dagger_{\mathbf{q}}] \tag{22}$$

Note that the second term vanishes due to $\hat{a}$ and $\hat{b}$ commuting. We are therefore left with:

$$[\hat{Q}, \hat{a}^\dagger_{\mathbf{q}}] = \int \frac{d^3p}{(2\pi)^3} (2\pi)^3 \delta(\mathbf{p} - \mathbf{q}) \hat{a}^\dagger_{\mathbf{q}} = \hat{a}^\dagger_{\mathbf{q}} \tag{23}$$

We can now find $\hat{Q}\left|\mathbf{p}\right\rangle$:

$$\hat{Q}\left|\mathbf{p}\right\rangle_a = \hat{Q}\hat{a}^\dagger_{\mathbf{p}}\left|0\right\rangle = ([\hat{Q},\hat{a}^\dagger_{\mathbf{p}}] + \hat{a}^\dagger_{\mathbf{p}}\hat{Q})\left|0\right\rangle = \hat{a}^\dagger_{\mathbf{p}}\left|0\right\rangle \tag{24}$$

Since, logically the charge of the vacuum state $\hat{Q}\left|0\right\rangle = 0$, we are left with:

$$\hat{Q}\left|\mathbf{p}\right\rangle_a = \left|\mathbf{p}\right\rangle_a \tag{25}$$

Meaning the eigenvalue of $\hat{Q}$ on an a state is $+1$. Similar logic follows on a b state:

$$\hat{Q}\left|\mathbf{p}\right\rangle_b = -\left|\mathbf{p}\right\rangle \tag{26}$$

Implying that the quantised charge is reversed for a b state, its eigenvalue being -1.

Therefore, we find for a complex massive scalar field, there exists both particles and antiparticles, whose charges are quantised and opposite (and conserved.)